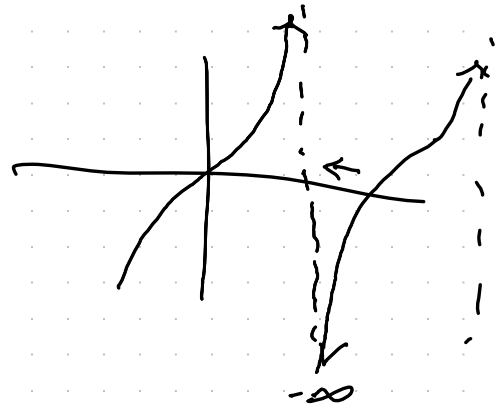
L'Hôpital examples – 0.4.8

Examples:

$$\lim_{x \to 0} \frac{e^x - 1}{\sin(x)} \overset{L'H}{=} \lim_{x\to 0} \frac{e^x}{\cos(x)} = \frac{1}{1} = 1$$

$$\lim_{x \to \frac{\pi}{2}^+} \frac{\ln(x - \frac{\pi}{2})}{\tan(x)} \quad \begin{matrix} \to -\infty \\ \to -\infty \end{matrix}$$

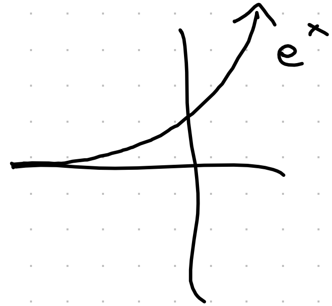


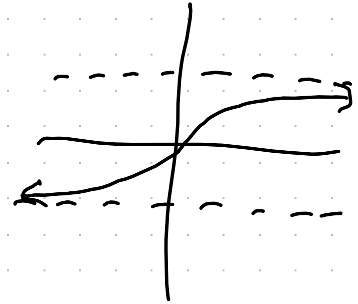
$$\overset{L'H}{=} \lim_{x \to \frac{\pi}{2}^+} \frac{\frac{1}{x - \frac{\pi}{2}}}{\sec^2(x)}$$

$$= \lim_{x \to \frac{\pi}{2}^+} \frac{\cos^2(x)}{x - \frac{\pi}{2}} \quad \begin{matrix} \to 0 \\ \to 0 \end{matrix}$$

$$\overset{L'H}{=} \lim_{x \to \frac{\pi}{2}^+} \frac{2\cos(x) \cdot -\sin(x)}{1} = -2\cos(\tfrac{\pi}{2})\sin(\tfrac{\pi}{2}) = 0$$

$$\lim_{x \to -\infty} \frac{e^x \;\to 0}{\arctan(x) + \frac{\pi}{2} \;\to 0}$$





$$\overset{L'H}{=} \lim_{x \to -\infty} \frac{e^x}{\frac{1}{1+x^2}}$$

$$= \lim_{x \to -\infty} \frac{1+x^2 \;\to \infty}{e^{-x} \;\to \infty} \quad \overset{L'H}{=} \lim_{x \to -\infty} \frac{2x \;\to -\infty}{e^{-x} \cdot -1 \;\to -\infty}$$

$$\overset{L'H}{=} \lim_{x \to -\infty} \frac{2}{e^{-x}} = 0$$

We can use L'Hôpital's rule to compare rates of growth.

<u>Example</u>:

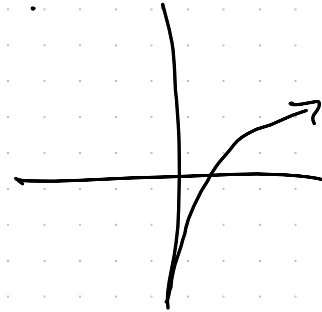
$$\lim_{x\to\infty} \frac{\overset{\nearrow\infty}{x^n}}{\underset{\to\infty}{e^x}} \overset{L'H}{=} \lim_{x\to\infty} \frac{\overset{\nearrow\infty}{n x^{n-1}}}{\underset{\to\infty}{e^x}}$$

$$\overset{L'H}{=} \lim_{x\to\infty} \frac{\overset{\nearrow\infty}{n(n-1)x^{n-2}}}{\underset{\to\infty}{e^x}} \overset{L'H}{=} \cdots \overset{L'H}{=} \lim_{x\to\infty} \frac{n(n-1)(n-2)\cdots(2)(1)}{e^x}$$

$$= \lim_{x\to\infty} \frac{n!}{e^x} = 0$$

e^x grows faster than any polynomial

Example: Comparing x^n and $\ln(x)$:



$$\lim_{x\to\infty} \frac{x^n \to \infty}{\ln(x) \to \infty}$$

$$\overset{L'H}{=} \lim_{x\to\infty} \frac{n x^{n-1}}{1/x} = \lim_{x\to\infty} n x^n = \infty$$

$\ln(x)$ grows slower than any polynomial

If we don't have $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$, L'Hôpital's rule usually fails, for instance

$$\lim_{x\to 1} \frac{x^2+5}{3x+4} = \frac{6}{7}, \quad \lim_{x\to 1} \frac{x^2+5}{3x+4} \neq \lim_{x\to 1} \frac{2x}{3} = 2$$